

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

May - June 2024

22MCA203 – OPERATING SYSTEMS WITH UNIX

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **One full question** from each Unit.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|--------------|------------|------------|------------|
| 1. a) List and explain the different services provided by the OS. | 08 | L*1 | 1 | 2 |
| b) What is multi-level queue and multi-level feedback queue? Highlight its advantages and disadvantages. | 08 | L2 | 1 | 1 |
| c) Explain the different states of a process with a neat diagram. | 04 | L1 | 1 | 2 |
| 2. a) Describe the implementation of IPC using shared memory & message passing. | 08 | L2 | 1 | 2 |
| b) What is a thread? List and explain the different multithreading models, bringing out the benefits of each. | 08 | L2 | 1 | 1 |
| c) Differentiate between: | | | | |
| i) Long-term and short-term scheduler, | | | | |
| ii) User level and kernel level threads. | 04 | L2 | 1 | 2 |

Unit – II

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|--|----|----|---|---|
| 3. a) What do you mean by race condition? Explain Reader's- Writer's problem with semaphore in detail. | 08 | L2 | 2 | 2 |
| b) Consider the following set of processes with the length of CPU burst time given in msec. Compute the average waiting time and average turnaround time for the below processes using SJF, Preemptive priority and Round Robin (Time quantum = 1msec) scheduling algorithm. | | | | |

Process	Arrival time	Burst time	Priority
P ₁	0	7	3
P ₂	3	2	2
P ₃	4	3	1
P ₄	4	1	1
P ₅	5	3	3

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|---|----|----|---|---|
| c) What are semaphores? Explain Binary and counting semaphores with an example. Explain the implementation of wait() and signal() semaphore operation. | 08 | L3 | 2 | 1 |
| 4. a) Consider the following set of processes with the length of CPU burst time given in msec. Compute the average waiting time and average turnaround time for the processes below using FCFS, Non-Preemptive priority, SRTF scheduling algorithm. | 04 | L2 | 2 | 2 |

Process	Arrival time	Burst time	Priority
P ₁	0	7	2
P ₂	2	2	3
P ₃	3	3	1
P ₄	4	1	1

- | | | | | |
|---|----|----|---|---|
| b) Explain the different scheduling criteria. Mention the cause of starvation and how to overcome it. | 08 | L3 | 2 | 1 |
| c) Explain Dining-Philosopher's problem using monitors. | 08 | L2 | 2 | 2 |
| | 04 | L2 | 2 | 2 |

P.T.O

Unit – III

5. a) With the help of a system model, explain deadlock & discuss the necessary conditions that must be held simultaneously in a system for a deadlock to occur. 08 L2 3 1
- b) Consider the following page reference string 7 0 1 2 0 3 0 4 2 3 0 3 2 3. Find out the number of page faults if there are 4-page frames, using the following page replacement algorithm i) Optimal Algorithm ii) FIFO Algorithm iii) LRU 12 L3 3 2
6. a) Explain with the help of supporting hardware diagram how the TLB improves the performance of a demand paging system. 08 L2 3 1
- b) Consider a Snapshot of the system at time T_0 :

Process	Allocation	Max	Available
	A B C D	A B C D	A B C D
P ₀	0 0 1 2	0 0 1 2	1 5 2 0
P ₁	1 0 0 0	1 7 5 0	
P ₂	1 3 5 4	2 3 5 6	
P ₃	0 6 3 2	0 6 5 2	
P ₄	0 0 1 4	0 6 5 6	

What is the content of need matrix?

Is the system in a safe state? If so, what is the safety sequence?

12 L3 3 1

Unit – IV

7. a) Explain the following basic UNIX commands with syntax and example.
 i) To List files and directories with all options
 ii) To Create a file and add data to it
 iii) To change file permission with its options
 iv) To print the current working directory
 v) To copy the output to another file
 vi) To open a string calculator
 vii) To get the calendar for June 2010
 viii) To know the current version of UNIX installed 08 L1 4 2
- b) Differentiate between hard links and symbolic links. 08 L2 4 2
- c) Explain the use of /dev/null and /dev/tty. 04 L1 4 2
8. a) Explain with examples absolute permissions and relative permissions assigned to a file. 08 L1 4 2
- b) Briefly discuss pattern matching with wild cards. Provide suitable examples for each. 08 L2 4 2
- c) Compare relative and absolute pathnames. 04 L1 4 2

Unit – V

9. a) Explain the difference between an internal command and an external command in the context of shell programming. Provide examples of each. 08 L1 5 2
- b) What are signals in the context of killing processes? Explain how you can use signals to terminate a process. 08 L2 5 2
- c) Describe the mechanism of process creation in Unix/Linux systems. What are the different ways to create a process? 04 L1 5 2
10. a) Define shell variables and environment variables. How are they different, and how can you manipulate them in shell scripts? 08 L1 5 2
- b) How do you run jobs in the background in a shell script? Provide an example demonstrating the use of background processes. 08 L2 5 2
- c) Explain the differences between the "at" and "batch" commands for scheduling tasks in Unix/Linux systems. 04 L1 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
